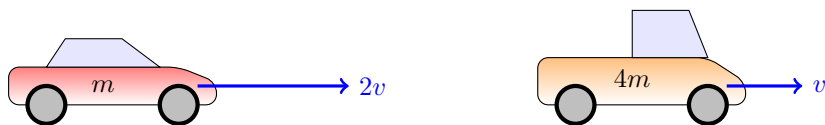


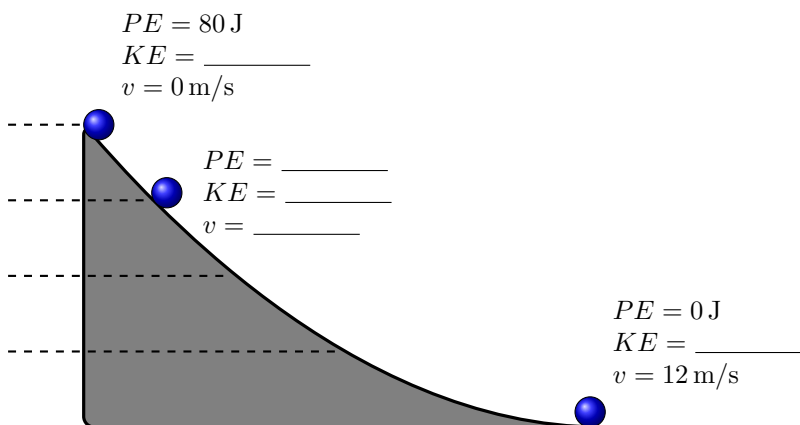
Tue, Feb 10 (Pd 1, 6) / Wed, Feb 11 (Pd 4)

Warmup

1. A car's speed is cut by $1/3$. By what factor does the car's kinetic energy change?
2. A car's speed is cut by $1/5$. By what factor does the car's kinetic energy change?
3. A car has twice the speed of a truck. However, the truck has four times as much mass. Which vehicle has a larger kinetic energy?



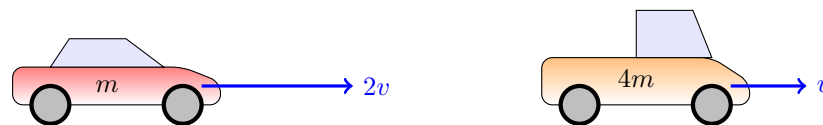
4. A marble starts at rest at the top of a frictionless ramp with 80 J of potential energy. At the bottom of the ramp, the ball is going 12 m/s. Use proportional reasoning to find the velocity of the ball when it is only $1/4$ of the way down the ramp. Use the image below to help you



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